Uniwersytet SWPS

Task list no. 9 – In this lab, we will implement the rod-cutting, matrix chain, and longest common subsequence problems using dynamic programming.

Task 1. Implement the DP algorithm for the rod-cutting problem using a non-recursive bottom-up approach, and find the revenue and the optimal cut for the following price list.
a) Run your code for the price list of {0, 2, 4, 8, 8, 10, 20, 21, 25, 28, 29}

Task 2. Implement the DP algorithm for the matrix chain problem, print the minimum number of multiplications, and the optimal parameterization.

Run your code for this dimension list for your matrix P = {10, 20,100,1,50}

Note. ALL PROGRAMS SHOULD COMPILE WITHOUT ERROR AND WORK (i.e. DO EXACTLY WHAT IS DESCRIBED IN THE TASK)!!!

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